

week 1

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Report: Tokenization and Sentiment Analysis using NLP

Objective

The objective of this task was to implement basic Natural Language Processing (NLP) techniques such as tokenization and stemming, and to build a sentiment analysis model using TF-IDF features. The task also involved exploring real-world NLP datasets using the Hugging Face `datasets` library.

Methodology

Tokenization and stemming were performed using the NLTK library. Tokenization splits text into individual words, while stemming reduces words to their root forms. Stopwords were removed to eliminate commonly occurring but less meaningful words. Due to updates in recent Python versions, an additional NLTK resource (`punkt_tab`) was required for successful tokenization. spaCy was also explored as an alternative tool, offering efficient tokenization and lemmatization.

For sentiment analysis, the IMDb movie review dataset was used. Text data was converted into numerical features using TF-IDF vectorization. A Logistic Regression classifier was trained on these features to predict sentiment labels.

Conclusion

The experiment demonstrated that classical NLP techniques combined with TF-IDF and machine learning classifiers can effectively perform sentiment analysis. This work provides a strong foundation for exploring advanced NLP models such as transformer-based architectures.